Practice workbook · Types of data

Distinguishing numerical from descriptive data and first-hand from existing data.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: PSY-RM-09

1. Question 4 · part c · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part c only.

4 A school is implementing a ban on unhealthy food.

Davide is interested to see whether young people would change their attitude towards eating unhealthy food due to the ban. 10 participants aged 11 to 16 years old from the school were asked to rate whether you should eat unhealthy food, using a scale of 1 to 10, with 1 being 'should never eat it' and 10 being 'should eat it every day'.

Davide tested the participants two months before the ban and again two months after the ban was put in place.

The results of Davide's study are shown in **Table 4**.

Participant	Attitude towards eating unhealthy food before the ban (1–10)	Attitude towards eating unhealthy food after the ban (1–10)
A	8	6
B	7	6
C	7	4
D	4	1
E	8	5
F	9	5
G	7	5
H	6	4
I	7	3
J	10	7

Table 4

(a) Calculate the mode for attitude towards eating unhealthy food **before** the ban.

(1)

Mode .....

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10

P 7 8 7 3 4 A 0 1 0 5 2

1. Question 4 · part c · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part c only.

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(b) Explain **one** conclusion that Davide could make about attitudes towards eating unhealthy food after the ban was put in place from the results in **Table 4**.

(2)

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(c) Davide decides to conduct further research gathering qualitative data.

Define what is meant by 'qualitative data'. Use an example in your answer.

(2)

(Total for Question 4 = 5 marks)

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1. Question 4 · part c · 2 marks · Working and answer

Working and answer

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2. Question 6 · part d · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part d only.

6 Kayla investigated whether the use of positive facial expressions (such as smiles) by actors in television advertisements resulted in people being more likely to buy the product being advertised.

First, she analysed two newspaper articles about the use of facial expressions in television advertisements.

Kayla then used an online questionnaire to find out if facial expressions influenced the purchase of a particular brand of cereal. 39,980,000 responses were collected in total.

(a) Give 39,980,000 in standard form.

(1)

39,980,000 in standard form .....

(b) 10,003,010 responses showed positive facial expressions used by actors in television advertisements did not influence their decision to buy the cereal.

Estimate the number of responses that showed positive facial expressions **did** influence their decision to buy the cereal.

(1)

Estimation .....

(c) 63% of the total responses to the online questionnaire were completed by families with children.

Calculate the number of families with children who completed the online questionnaire.

(1)

Number of families with children who completed the online questionnaire .....

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10

P 5 9 6 6 9 A 0 1 0 4 8

2. Question 6 · part d · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part d only.

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(d) Complete **Table 4** to show which data used by Kayla in her investigation was primary data and which was secondary data.

(2)

Type of data	Example from Kayla's investigation
Primary data	
Secondary data	

Table 4

(Total for Question 6 = 5 marks)

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Working and answer

3. Question 1 · part b · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part b only.

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1 Zak is investigating internet addiction. He designs a questionnaire that includes the following question:

How often do you use social media? Please tick one answer.

☐ Less than five times each day.
☐ Between five and ten times each day.
☐ More than ten times each day.

Figure 1

(a) Identify the type of question shown in **Figure 1**.

(1)

☒ **A** Unstructured question

☒ **B** Open-ended question

☒ **C** Closed-ended question

☒ **D** Semi-structured question

(b) State the type of data Zak will gather from his question in **Figure 1**.

(1)

.....

.....

(c) Zak found that nine out of fifteen participants used social media more than ten times each day.

Calculate the number of participants who used social media more than ten times each day as a fraction of all participants.

You must give your answer in the lowest form.

(1)

Fraction .....

(Total for Question 1 = 3 marks)

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3. Question 1 · part b · 1 mark · Working and answer

Working and answer

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4. Question 1 · part a · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a only.

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1 Andrew conducted a laboratory experiment to investigate the capacity of short-term memory. He showed participants 20 trigrams and asked them to recall as many of the trigrams as they could.

(a) Andrew counted how many trigrams each participant recalled correctly.

Identify the type of data gathered by Andrew.

(1)

- ☒ **A** Qualitative data
- ☒ **B** Subjective data
- ☒ **C** Quantitative data
- ☒ **D** Secondary data

(b) Andrew used a standardised procedure when showing the trigrams to his participants.

Give **two** ways that Andrew could have standardised his procedure.

(2)

1 .....

.....

.....

2 .....

.....

.....

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4. Question 1 · part a · 1 mark · Working and answer

Working and answer

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5. Question 3 · part c · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part c only.

3 Juliet investigated whether there was a relationship between the number of hours spent using social media and self-rated body image scores.

Firstly, Juliet interviewed a sample of 17-year-old students using a structured interview method, where she asked them to state the length of time they spent on social media per day.

Juliet then asked the same participants to complete a questionnaire to rate how happy they were with their appearance, with a score of 0 being 'very unhappy' and 10 being 'very happy'.

(a) Define what is meant by a 'structured interview'.

(1)

(b) Describe how Juliet could have used a random sampling technique for her investigation.

(2)

(c) Identify the type of data Juliet gathered in her investigation.

(1)

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8

P 7 0 4 8 6 A 0 8 4 8

5. Question 3 · part c · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part c only.

The results of Juliet's investigation are shown in **Table 3**.

Participant	Number of hours per day using social media	Body image score (out of 10)
A	2	6
B	1	7
C	4	3
D	6	2
E	2	7
F	5	3
G	1	9
H	2	7
I	7	1

Table 3

(d) Calculate the mode for body image scores.

(1)

Mode .....

(e) Juliet concludes that there is a negative relationship between social media use and body image.

Give **one** way Juliet's findings could support this conclusion.

(1)

.....

.....

.....

(Total for Question 3 = 6 marks)

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5. Question 3 · part c · 1 mark · Working and answer

Working and answer

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6. Question 2 · part c · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c only.

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2 Jason is planning to use a questionnaire to investigate whether there is a difference in the use of social media by people of different ages.

(a) Give **one** open-ended question and **one** closed-ended question that Jason could use in his questionnaire.

(2)

Open-ended question

.....

.....

.....

Closed-ended question

.....

.....

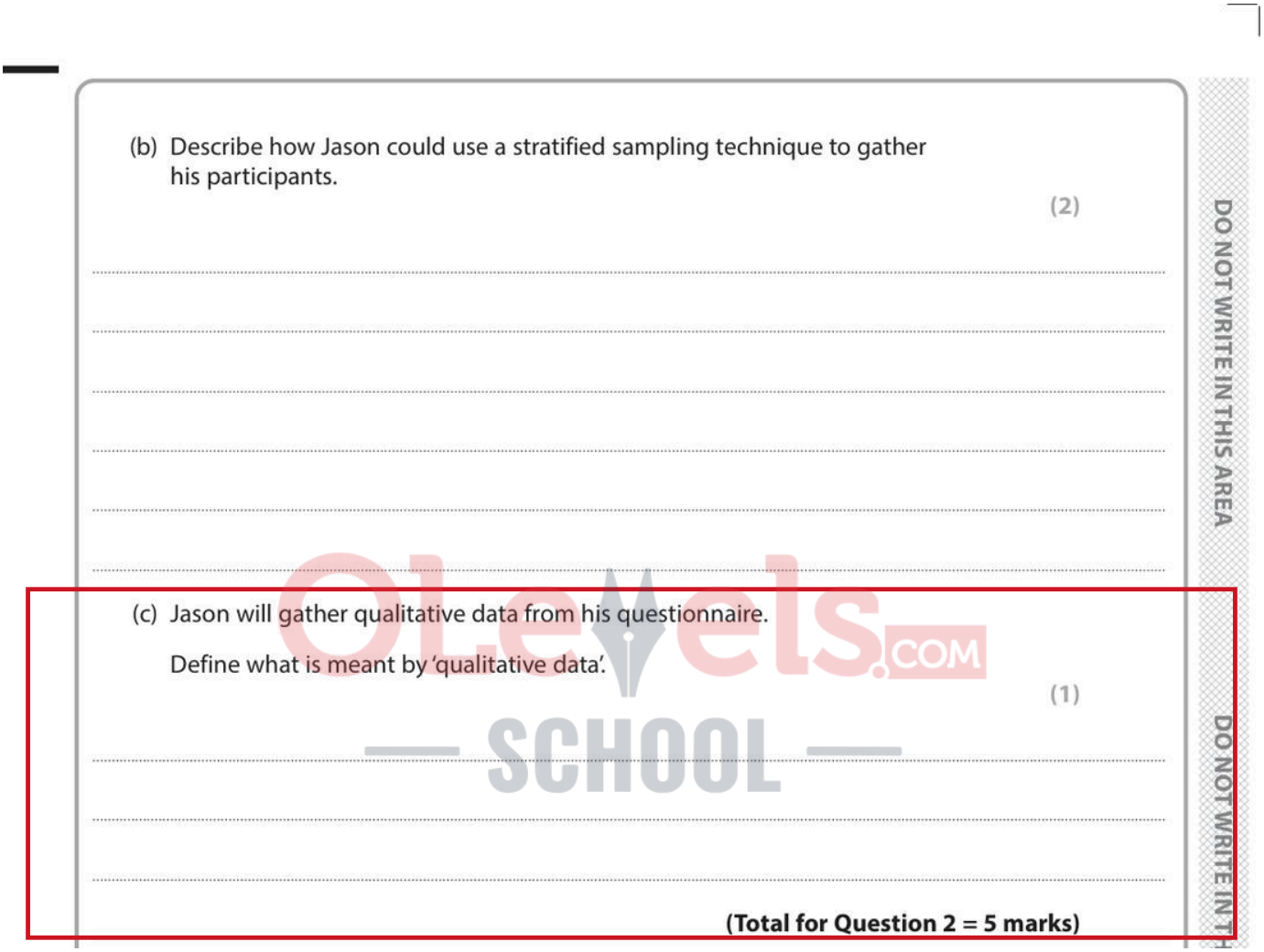
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6. Question 2 · part c · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c only.



(b) Describe how Jason could use a stratified sampling technique to gather his participants.

(2)

(c) Jason will gather qualitative data from his questionnaire.

Define what is meant by 'qualitative data'.

(1)

(Total for Question 2 = 5 marks)

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Working and answer

7. Question 5 · part b · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

5 Rocio conducted a case study on her patient who had an addiction to alcohol. She worked with the patient for six months to help treat their addiction.

(a) Define what is meant by a 'case study'.

(1)

(b) Rocio recorded the conversations with the patient as part of her case study. She also monitored the alcohol consumption of the patient in units of alcohol (one unit of alcohol = 10 ml).

Her results are shown in **Table 2**.

Month	Total alcohol consumption in units
March	298
April	221
May	181
June	140
July	99
August	59

Table 2

Describe, using an example from Rocio's case study, what is meant by 'quantitative data'.

(2)

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7. Question 5 · part b · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

(c) Estimate the number of units of alcohol that the patient has reduced their consumption by between March and August.

(1)

Estimation .....

(d) For the months of March and April, Rocio recorded the patient's alcohol consumption each week. Her results are shown in **Table 3**.

Week	Total alcohol consumption in units
March Week 1	79
March Week 2	77
March Week 3	72
March Week 4	70
April Week 1	79
April Week 2	63
April Week 3	45
April Week 4	34

Table 3

Calculate the mode for the units of alcohol consumed by the patient using **Table 3**.

(1)

Mode .....

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7. Question 5 · part b · 2 marks · Working and answer

Working and answer

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